

Seasonality in Commodities Market and Generalized Seasonality Test via Sequential Regression Testing of Prime Cycle Periodicity

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1 2 3

Abstract

The practice of seasonality investing in commodities market is discussed in this paper. We review performance of classical seasonality portfolios, and implement a novel generalized seasonality test developed by Shanaev and Ghimire [1] for cryptocurrency and stock markets on commodity futures. Python code implementing sequential dummy variable regressions for prime cycle periodicity as hypothesis test for existence of seasonal patterns against no seasonality with FER control is presented.

0.1 A Note of Precaution

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1 Introduction

Seasonality investing is a well-known study with extensive literature on identifying cyclicalities in the returns of assets. Some of these effects include the ‘Sell in May and go away’ hypothesis, January effect, Friday theta pricing effect and a host of other calendar effects. Not unlike the class of behavioral finance literature used to attribute excess returns, with every cyclical effect, there is an accompanying, unverifiable fundamental conjecture. The slew of tests that different quantitative researchers subject time-series to in accordance to each hypothesized calendar effect is not adjusted for multiple testing, and taken as a whole the familywise error rate (FER) of seasonality literature must overestimate the presence of true seasonality. Therefore, the criticism of financial literature towards seasonality literature being congruent with data-mining is not unfounded. The work of Shanaev and Ghimire [1] develops a novel generalized seasonality test using sequential dummy variable regressions for prime cycle periodicity as hypothesis test for existence of any seasonal patterns against the broad null hypothesis of no seasonality. A harmonic mean p-value is computed to ensure FER control for multiple testing.

In this paper, a classical, vanilla portfolio is created using a walkforward test to generate a long-short portfolio. The prime cycle generalized seasonality test is discussed and its ability to identify time-series seasonality is shown. This novel approach allows us to quantify in probabilistic terms the strength of observed seasonality, and has applications in both econometric problems and practical concerns such as the construction of a seasonality investment portfolio. Code and discussions are included.

2 Classical Seasonality Portfolio

A seasonality portfolio for commodities can typically be constructed by looking at harvest and non-harvesting seasons. Corresponding to the futures contract month, new/old crop months determine the cyclicalities of the futures commodities price. For instance, the seasonality of grains on the CME website⁴, and a popular seasonality investment research database⁵ shows that returns vary non-

⁴<https://www.cmegroup.com/education/courses/introduction-to-grains-and-oilseeds/understanding-seasonality-in-grains.html>

⁵<http://www.equityclock.com/>

trivially based on the time of the year. Of course, for all practical concerns, these price movements must be systematically under or over estimated in the commodity term structure for there to be excess speculative returns in futures markets. A general quantitative seasonality long/short portfolio may be constructed by the simple process of bucketing returns over a historical calibration period, and going long/short the best x/y periods.

To demonstrate the performance of such a strategy, the following methodology is constructed: from historical futures data, create a continuous price series. To keep the price behavior fairly well behaved, the price series is constructed from back month (second contract to maturity), and is rolled over to the far month (third contract to maturity) as the front month expires (and the far month becomes the back month contract). As a result, we will not be subject to idiosyncratic movements in the front month contract, which is often subject to price manipulation and other exogenous effects. In addition, the rollover is conducted as a linear weighted roll over a ten day span (see method on QuantStart)⁶ from the back month to the far month contract. For simplicity, settlement prices are used for pnl computation. Necessarily, these implementations assume fractional holdings, tax-free investments, no active trading fees, spread, market impact or slippage. This further implies sufficient liquidity for the entirety of the backtest period for the contracts traded. In walk-forward fashion, five years of historical returns are used to calibrate positions. Specifically, at the beginning of each year, the previous five years worth of monthly returns are bucketed and sorted, and a time-series directionally neutral portfolio is constructed to go long the best three months and short the worst three months. Results in comparison to the buy-and-hold of the perpetual series are presented (labelled with Sharpe) for CJ (cocoa), CL (crude oil), HE (lean hog), HO (heating oil), NG (henry hub natgas), RB (rbob gasoline), ZC (corn), ZS (soybean) and ZW (wheat) with futures data from the CME. The portfolio returns and pnl plots are presented in Appendix A. Notably, a first-glance suggests potential seasonality effects in HE, NG, RB and ZW, with HE (heating oil) having the clearest seasonality outperformance over the buy-and-hold portfolio. However, from the pnl graphs, no probabilistic statements may be made about the seasonality effect and statistical significance for purpose of seasonality portfolio construction. We investigate this problem in the next section.

⁶<https://www.quantstart.com/articles/Continuous-Futures-Contracts-for-Backtesting-Purposes/>

3 Generalized Seasonality Test

The existence of any seasonal pattern against the broad null hypothesis of no seasonality is constructed. For time series of n observations, some maximum period $m \leq n$, let the number of primes be $k = |\pi(m)|$, where $\pi(m)$ is the list of primes $\leq m$. For instance, when $m = 8$, then $\pi(m) = [2, 3, 5, 7]$ and $k = 4$. The regression analysis tests for periodicity effects in the prime cycles of $\pi(m)$ - note that this in turn specify the periodicity effect present in any index n that can be represented as a product of the primes in $\pi(m)$. That is, again for $m = 8$, (corresponding to daily granularities) $2 \times 2 \times 3 \times 3 \times 7 = 252$ can be used to test for the presence of a ‘day of year’ effects, while (corresponding to monthly granularities) 3 can be used to test for ‘month of quarter’ effects, and $2 \times 2 \times 3$ for ‘month of year’ effects. The regression test is conducted as follows: first create a set of dummies for $i \leq n, j \leq k$ constructed by the procedure $X_{ij} = \mathbb{1}\{i \bmod \pi(m)_j = 0\}$ - that is, the j -th dummy variable has a one-entry when the i -th index is divisible by prime $\pi(m)_j$ or zero otherwise. For reference and code, see Appendix B. Note that by experimental design, the multicollinearity of the regression test is reduced - there are no high frequency correlations of any group of dummy variables, since they are jointly set to one only at the frequency of their lowest common multiples. This regression test is fitted to the return series sequentially for $j : 1 \rightarrow k = |\pi(m)|$, where the j -th regression tests up to and including the j -th prime. Each regression test is accompanied by an F-test for the joint significance of prime cycle returns where the F-statistic $\sim F_{j, n-j-1}$. A total of k -regressions with k F-test p-values are then adjusted for multiple testing using the harmonic mean to give $p_{adj} = \frac{k}{\sum_{i=1}^k \frac{1}{p_i}}$. When the adjusted p-value is lower than an alpha value (say, $\alpha = 0.05$), then the null hypothesis of no seasonality is rejected in favour of the alternative hypothesis. For a significant regression test, the significance of individual t-tests for the beta coefficients and their signs for dummy variable $X_{.,j}$ indicate the relation between asset returns and the prime cycle periodicity of duration $\pi(m)_j$ at holding period of specified granularity.

4 Results

A test is constructed for the existence of seasonality with holding period of one month, and used to survey the walkforward portfolios discussed in Section 2. The results of the sequential regression tests are presented in Table 1.

Market	p_{adj}	β_2	β_3	β_5	2M	3M	5M.
CJ	0.224	0.009	0.009	-0.006	0.184	0.245	0.532
CL	0.65	-0.006	-0.003	-0.011	0.53	0.773	0.306
HE	0.023	0.002	0.023	0.004	0.765	0.003	0.68
HO	0.374	0.008	0.009	0.001	0.303	0.299	0.902
NG	0.793	0.003	0.011	0.009	0.811	0.46	0.606
RB	0.482	0.012	0.008	0.01	0.353	0.528	0.533
ZC	0.43	-0.002	-0.005	-0.012	0.75	0.384	0.068
ZS	0.622	0.003	-0.006	0.004	0.606	0.356	0.588
ZW	<i>0.133</i>	-0.008	-0.008	0.003	<i>0.119</i>	<i>0.158</i>	0.612

Table 1: Harmonic Mean Adjusted p-valued at the 2, 3, 5-month prime cycle periodicity

The p-values ≤ 0.10 are **bolded** and p-values ≤ 0.15 (rounded-down) are *italicized*. Some comments are made.

The HE (lean hogs) time-series show statistical significance on both the overall adjusted p-value and the 3M periodicity, suggesting the presence of month of quarter effects. This is not corroborated by significance in either the 2M or 5M cycles, suggesting that the $2 \times 3 = 6M$ half-year cycles are not significant. Additionally, there are no irregular periodicities observed (the 5M prime cycle do not correspond to any meaningful calendar effects, except perhaps a long-term cycle at $2 \times 2 \times 3 \times 5 = 60M = 5Y$ cycle). This is reflected in the strong excess returns of the HE lean hogs portfolio in the walkforward (see Appendix A, HE). On the other hand, the 5M prime cycle for ZC appears to be statistically significant, but when controlled for FER, the adjusted p-value is 0.43, suggesting no statistical significance. The poor performance of the seasonality portfolio of the ZC walkforward agrees (see Appendix A, ZC). Notably, ZW has a moderately low p-value in the seasonality test, with an adjusted p-value of 0.133. This is matched with a moderately low p-value at the 2M and 3M cycle (0.119, 0.158) respectively, suggesting half-year cycles and possibly quarterly cycles. As noted, the ZW walkforward portfolio (see Appendix A, ZW) also performed well in the walkforward. Despite the decent performance of the NG and RB walkforward portfolio, both the adjusted p-value and individual t-tests conclude no significance under the generalized

seasonality test.

5 Discussions, Code and In-Sample Portfolio Construction

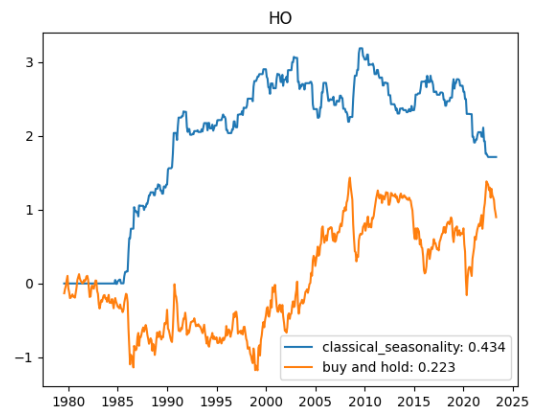
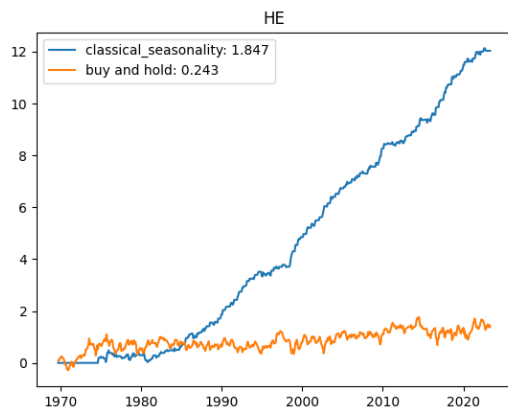
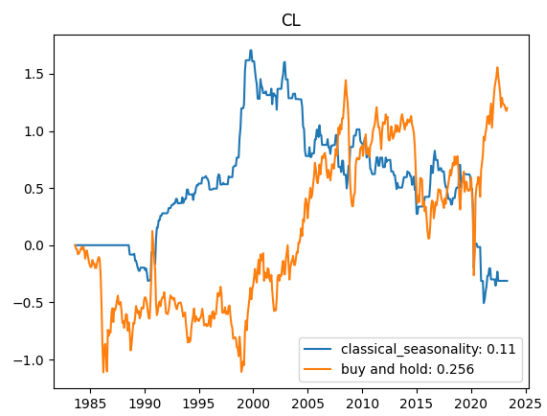
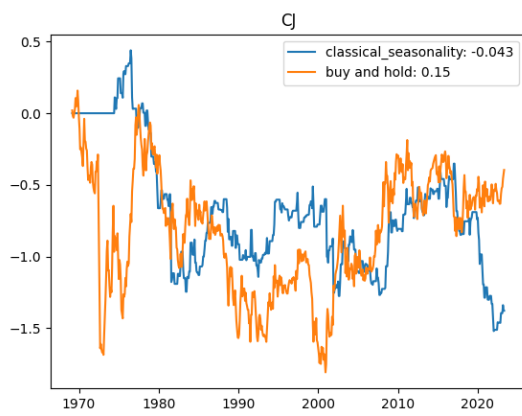
Our quantitative experiment demonstrated the ability to identify seasonality in the HE and ZW market, while identifying the prime periodicities and durations for such effects based on the joint significance of primes. The holding time for the seasonality portfolio can be tested for by using different granularity candles on the time axis. Although NG, RB were profitable in the walkforward portfolio, the hypothesis testing for seasonality in these instruments showed no statistical significance. Possibly, this is because the NG and RB markets displayed persistent but dynamic periods of local seasonality, which was picked up in the walkforward analysis but not in the global seasonality test.

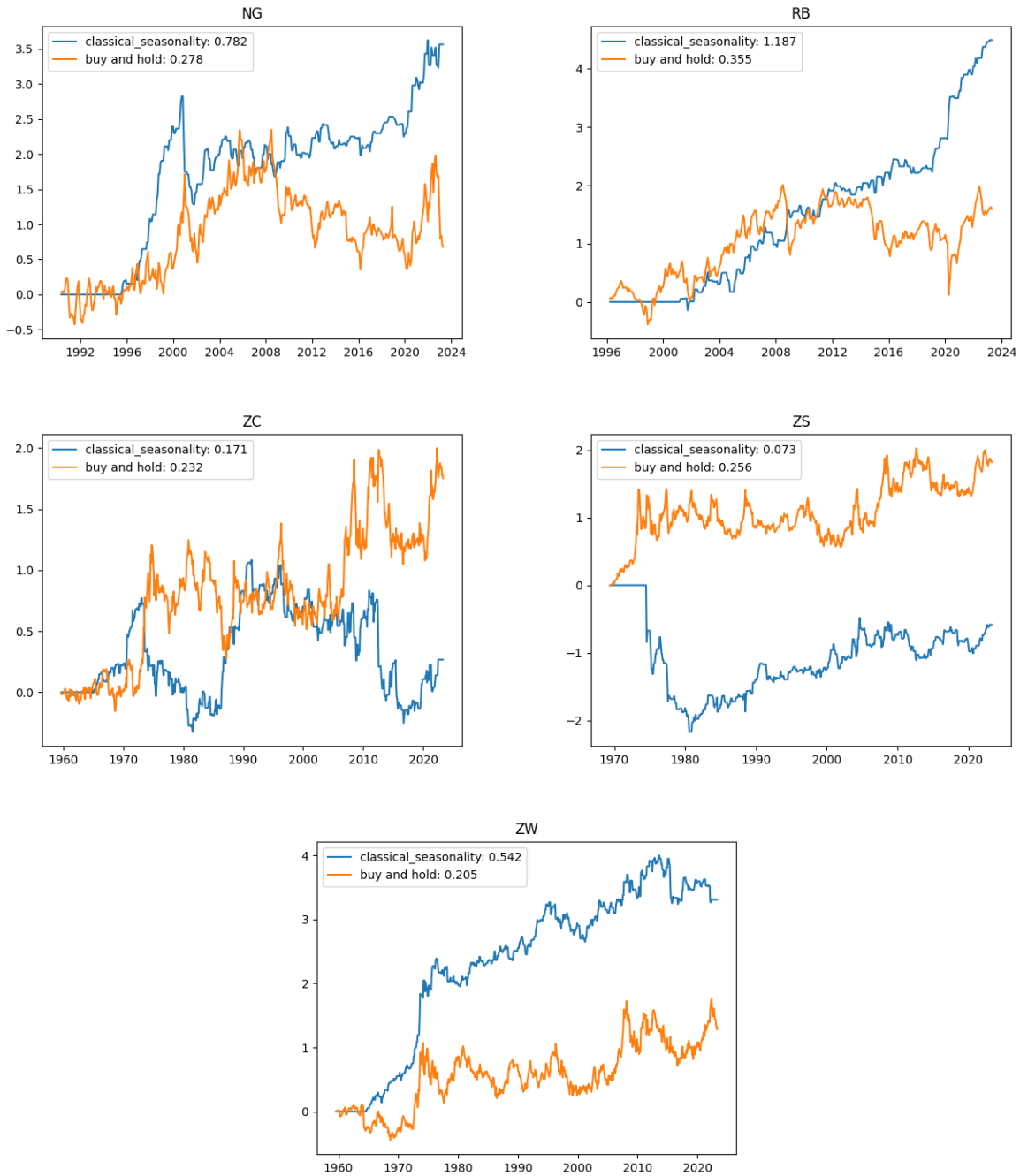
The generalizability of the test gives us a powerful toolkit for assessing the efficiency of a financial time series. In particular, the seasonality hypothesis test does not assume any priors on the structure or duration of the calendar effects, potentially enabling the investor to discover cyclicity deviating from the standard notes made by quantitative researchers. Additionally, the harmonic p-value adjustment for the familywise error rate gives us a better revision of the data-snooping bias present when the same time-series is subjected to a battery of tests for calendar effects.

For practitioners of seasonality investing, the applications are many. For instance, the proposed test can be used to determine the inclusion of a futures contract in a seasonality portfolio. Since this test only requires the presence of a univariate time-series of returns, a portfolio of cyclical stocks displaying seasonality may also be constructed. Alternatively, one may also directly construct an investment portfolio based on the regression model. On the other hand, the regression test may be used for generating seasonality-adjusted data, and can be purposed as part of a solution for many econometric problems.

To demonstrate the sequential regression testing implementation in Python, an in-sample seasonality portfolio is constructed together with the walkforward portfolio and is presented in Appendix B.

A Returns on the Vanilla Walkforward





B Regression Test Implementation

The following Python code generates the dummy prime feature matrix for the regression test:

```
1 def get_primes(n):
```



```

2     #https://stackoverflow.com/questions/2068372/fastest-way-to-list-all-primes-
    below-n
3     """ Returns a list of primes < n """
4     sieve = [True] * n
5     for i in range(3,int(n**0.5)+1,2):
6         if sieve[i]:
7             sieve[i*i::2*i]=[False]*((n-i*i-1)//(2*i)+1)
8     return [2] + [i for i in range(3,n,2) if sieve[i]]
9
10 def _get_feature_matrix(m,n):
11     primes=get_primes(m+1)
12     k=len(primes)
13     dummies=[]
14     for xk in range(k):
15         arr=np.zeros(n)
16         arr[primes[xk]-1::primes[xk]] = 1
17         dummies.append(arr)
18     X = np.vstack(dummies).T
19     return X,primes,k

```

with the following output for reference:

```

>> print(_get_feature_matrix(m=8,n=16))
<<
(array([[0., 0., 0., 0.],
       [1., 0., 0., 0.],
       [0., 1., 0., 0.],
       [1., 0., 0., 0.],
       [0., 0., 1., 0.],
       [1., 1., 0., 0.],
       [0., 0., 0., 1.],
       [1., 0., 0., 0.],
       [0., 1., 0., 0.],
       [1., 0., 1., 0.],
       [0., 0., 0., 0.]])

```

```

[1., 1., 0., 0.],
[0., 0., 0., 0.],
[1., 0., 0., 1.],
[0., 1., 1., 0.],
[1., 0., 0., 0.])), [2, 3, 5, 7], 4)

```

In the following script, a univariate time series is assumed to be arranged in a *pandas* dataframe with datetime index and column named *cont_ret*. For instance:

```

>> contract_df
      cont_ret
date
1969-02-13  0.027375
1969-02-14 -0.004796
1969-02-17 -0.013922
1969-02-18 -0.005430
1969-02-19  0.007917
...         ...
2023-04-17  0.003452
2023-04-18  0.015824
2023-04-19  0.003048
2023-04-20 -0.006077
2023-04-21  0.009171

```

The implementation of the generalized seasonality test, in-sample portfolio construction and walkforward portfolios are presented:

```

1 import re
2 import pytz
3 import numpy as np
4 import pandas as pd
5 import numpy_ext as npe
6 import matplotlib.pyplot as plt
7 import scipy.stats as scist

```

```

8 import statsmodels.formula.api as smf
9
10 from pprint import pprint
11 from datetime import datetime
12 from datetime import timedelta
13
14 contract, contract_df = "CL", pd.DataFrame() #retrieve data
15 mgrps=contract_df.groupby(by=[contract_df.index.month, contract_df.index.year])
16 monthlies=pd.concat([(1+mgrps.get_group(group).cont_ret).cumprod().tail(1)-1 for
    group in mgrps.groups])
17 monthlies=monthlies.loc[sorted(monthlies.index)]
18
19 '''global in-sample seasonality test and portfolio on monthly returns
20 '''
21 X,primes,k=_get_feature_matrix(m=5,n=len(monthlies))
22 kwargs={"X_"+str(i):arr for i,arr in zip(range(k),X.T)}
23 df=pd.DataFrame(index=monthlies.index, data={"r":monthlies.values, **kwargs})
24 multiple_tests={}
25 '''sequential regression testing'''
26 for j in range(k):
27     formula=f"r ~ {' + '.join(['C(' + str(kwarg) + ') ' for kwarg in kwargs.keys()
    if int(kwarg.split('_')[1]) <= j])}"
28     res=smf.ols(formula=formula,data=df).fit()
29     j_test={
30         "f_test": res.f_pvalue,
31         "coeffs": res.params,
32         "t_tests": res.pvalues
33     }
34     multiple_tests[j] = j_test
35
36 fpvals=[multiple_tests[j]["f_test"] for j in range(k)]
37 adj_pvals=k/np.sum([1/p for p in fpvals])
38 coeff_ps=multiple_tests[k-1]["t_tests"].values[1:]
39 coeff_vals=multiple_tests[k-1]["coeffs"].values[1:]
40 alpha=1 #ignore to show all in sample graphs
41
42 '''in-sample portfolio construction using the regression model'''

```

```

43 seasons=np.multiply(np.where(coeff_ps<alpha,1,0),coeff_vals)
44 votess = np.array([np.multiply(x,seasons) for x in X])
45 positions=np.array([np.sum(votes) for votes in votess])
46 print(contract)
47 print(round(adj_pvals,3))
48 print([round(p,3) for p in coeff_ps])
49 print([round(beta,3) for beta in coeff_vals], "\n")
50 df["positions"]=positions
51 df["insamp_ret"] = df.r*df.positions
52 plt.plot(np.log((1+df.insamp_ret).cumprod()),label="insamp_monthlies")
53 plt.title(contract)
54 plt.show()
55 plt.close()
56
57 '''walkforward classical seasonality test
58 based on last 5 years of data, long top 3 months and short bottom 3 months for the
59     next one year period
60 '''
61 mgrps=contract_df.groupby(by=[contract_df.index.month, contract_df.index.year])
62 classical_df=pd.concat([(1+mgrps.get_group(group).cont_ret).cumprod().tail(1)-1
63     for group in mgrps.groups])
64 classical_df=classical_df.loc[sorted(classical_df.index)]
65 classical_df=pd.DataFrame(index=classical_df.index,data={"date": classical_df.
66     index, "cont_ret": classical_df.values})
67 classical_df["positions"]=np.nan
68 def classical_roll(dt,r):
69     df=pd.DataFrame(index=dt,data={"r":r})
70     if not np.isnan(classical_df.at[dt[12*5],"positions"]):
71         return
72     calibrate_df=df.loc[:dt[12*5]] #first 5 year positions
73     mnth_grps=calibrate_df.groupby(by=calibrate_df.index.month)
74     mnthlies={grp: np.mean(mnth_grps.get_group(grp).r.values) for grp in mnth_grps
75         .groups}
76     mnthlies={k:v for k,v in sorted(mnthlies.items(), key=lambda pair: pair[1]) }
77     short,long=list(mnthlies.keys())[:3],list(mnthlies.keys())[-3:]
78     positions=[-1 if i.month in short else (1 if i.month in long else 0) for i in
79         dt[12*5:]] #next one year positions

```

```

75     classical_df["positions"].loc[dt[12*5:]] = positions
76
77     if len(classical_df)>12*6:
78         rollop=npe.rolling_apply(
79             classical_roll,
80             12*6,#5y calibration, 1 year walk
81             *classical_df[["date","cont_ret"]].T.values
82         )
83     classical_df=classical_df.fillna(0.0)
84     classical_df["classical_portfolio"]=classical_df.positions*classical_df.
cont_ret
85     ret_bnh=classical_df.cont_ret.loc[classical_df.cont_ret != 0].values
86     ret_sea=classical_df.classical_portfolio.loc[classical_df.classical_portfolio
!= 0].values
87     shrp_sea=round(np.mean(ret_sea)/np.std(ret_sea)*np.sqrt(12),3)
88     shrp_bnh=round(np.mean(ret_bnh)/np.std(ret_bnh)*np.sqrt(12),3)
89
90     plt.plot(np.log((1+classical_df.classical_portfolio).cumprod()),label=f"
classical_seasonality: {shrp_sea}")
91     plt.plot(np.log((1+classical_df.cont_ret).cumprod()),label=f"buy and hold: {
shrp_bnh}")
92     plt.title(contract)
93     plt.legend()
94     plt.show()
95     plt.close()

```

References

- [1] S. Shanaev and B. Ghimire. A generalised seasonality test and applications for cryptocurrency and stock market seasonality . https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3722154.

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